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POSTER

# how sleep consolidates memory — Poster

A Brain Health learning product

Foundational • General public

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Edition 1

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QRU Poster - Neuroscience

# how sleep consolidates memory - Poster - Free Preview

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# How Sleep Consolidates Memory

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## Sleep Is Your Brain's Nightly "Save, Sort, and Strengthen" System

When you learn something new, that memory is still fragile. During sleep, your brain reactivates, organizes, connects, and stabilizes what you learned - helping facts, skills, experiences, and emotions become easier to remember and use later.

Memorable takeaway:
Study plants the seed. Sleep helps it take root.

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## Visual Pathway: How Learning Becomes Stronger

Learn ? Hippocampus Quick Capture ? Deep Sleep Replay ? Neocortex Integration ? Stronger Recall

Use this as a simple pathway graphic for the poster:

1. Learn
You study, practice, read, listen, experience, or solve a problem.

2. Hippocampus Quick Capture
The hippocampus helps rapidly record new information - like a temporary "quick save."

3. Deep Sleep Replay
During slow-wave sleep, patterns of brain activity linked to recent learning can be reactivated.

4. Neocortex Integration
The neocortex gradually organizes new learning into broader knowledge networks.

5. Stronger Recall
Over time, memories may become more stable, connected, and easier to access.

## Important Clarification: Memory Is Not Simply "Moved" Like a File

Memory consolidation is not a simple physical transfer from the hippocampus to the neocortex.

Instead, it is a gradual process of reorganization and integration. The hippocampus and neocortex communicate through coordinated brain activity, helping new learning become linked with older knowledge, patterns, meanings, and experiences.

Simple summary:
Hippocampus = quick capture and early coordination
Neocortex = long-term integration and organized knowledge

## What Happens During Deep Sleep?

Deep Sleep Supports Stability, Facts, and Events

Deep sleep, also called slow-wave sleep, appears especially important for strengthening many kinds of declarative memory - including facts, vocabulary, concepts, and personal experiences.

During deep sleep, the brain may support:

- Memory replay: recently learned information is reactivated
- Brain network coordination: the hippocampus and neocortex communicate in organized rhythms
- Stronger useful pathways: important learning connections become easier to reuse
- Noise reduction: less useful activity may be reduced so the brain can work more efficiently
- Readiness for tomorrow: the brain becomes better prepared for new learning

## Continue Learning in the Full Edition

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8 more section(s) are included in the complete Treasure Standard(TM) edition.

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